Business Requirements Document (BRD)

Project Title: Expense Tracker with Smart Insighter

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1. Project Overview

The Expense Tracker with Smart Insighter is a smart financial management system that helps users track and manage their daily expenses in an efficient way.

Unlike traditional expense trackers, this system not only records transactions but also provides intelligent insights using AI techniques. It automatically categorizes expenses using Natural Language Processing (NLP) and reduces manual effort.

The system also includes a dashboard with charts and graphs to show spending patterns clearly. It can predict future expenses and alert users about possible overspending. In addition, it detects unusual transactions to help users stay aware of their financial activities.

Overall, this project aims to make expense tracking simple, smart, and more useful for better financial decision-making.

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2. Objectives

- To automate expense tracking and categorization
- To provide intelligent financial insights
- To predict future expenses and balances
- To alert users about overspending
- To improve financial planning and savings

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3. Scope of the Project

In Scope

- Expense tracking (manual entry)
- AI-based categorization
- Predictive analysis and alerts
- Anomaly detection
- Dashboard with visual analytics

Out of Scope

- Bank API integration
- Investment tracking
- Tax calculation system

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4. Key Features

4.1 AI Categorization

- Automatically categorizes expenses using NLP
- Reduces manual effort
- Improves accuracy over time

4.2 Predictive Alerts

- Forecasts end-of-month balance
- Alerts user about overspending
- Helps in better budgeting

4.3 Anomaly Detection

- Detects unusual or abnormal transactions
- Identifies sudden spending spikes
- Flags recurring subscription increases

4.4 Visual Analytics

- Provides real-time dashboards
- Displays charts and graphs
- Uses dynamic spending heatmaps

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5. Functional Requirements

5.1 User Module

- User registration and login
- Profile management

5.2 Expense Module

- Add, edit, delete expenses
- Store transaction data

- View and filter expenses

5.3 AI Module

- NLP-based categorization
- Spending prediction system
- Anomaly detection engine

5.4 Analytics Module

- Dashboard with charts
- Monthly and category-wise reports

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6. Non-Functional Requirements

- System should be fast and responsive
- Data should be secure
- User interface should be simple and easy
- System should handle multiple users

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7. Tech Focus

7.1 Full-Stack

- Frontend: React / Next.js
- Backend: FastAPI

7.2 Intelligence

- Python (Scikit-learn) for trend forecasting and clustering

7.3 Data

- PostgreSQL for secure and structured data storage

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8. Data Requirements

- User data (name, email, password)
- Expense data (amount, category, date)
- Transaction details (description, payment method)

9. User Flow

1. User logs into the system
2. User adds expense
3. AI categorizes the expense
4. Data is stored in database
5. Dashboard updates in real-time
6. System predicts future spending
7. Alerts and insights are shown

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10. Stakeholders

- End Users
- Developers
- Project Guide

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11. Success Criteria

- Accurate expense categorization
- Correct prediction of spending
- Fast dashboard performance
- Useful alerts and insights

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12. Future Enhancements

- Bank integration
- Mobile application
- Voice-based input
- Advanced AI models

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13. Conclusion

The Expense Tracker with Smart Insights is a smart financial system that tracks expenses and provides useful insights using AI.

It includes features like automatic categorization, prediction, anomaly detection, and dashboards to help users understand their spending and make better financial decisions.

Overall, it simplifies expense management and helps improve financial planning and savings.